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A machine learning model for the prediction of hessian matrices for two to four atomic molecules.

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Research Internship Report

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List of abbreviations

APF	Atom Pair Focused
BO	Born–Oppenheimer
CS	Coordinate System
ETR	Extra Tree Regression
ES	Electrostatic
LDA	Local Density Approximation
MI	Main Inertial
ML	Machine Learning
RFR	Random Forest Regression
XC	Exchange and Correlation

1 Aim

2 Motivation

3 Theory

3.1 Euler Angles

One method of describing coordinate system transformations, are the Euler angles. They serve as a mathematical tool to connect one CS with another.
The initial coordinate system $\mathbf{x}, \mathbf{y}, \mathbf{z}$, shown in Fig. 3.1 can be transformed into the $\mathbf{x}', \mathbf{y}', \mathbf{z}'$ CS by three rotations. The three Euler angles α, β and γ are defined as follows. The distance between the vectors $\mathbf{L}_\mathrm{a}$ and $\mathbf{L}_\mathrm{b}$ is shown in red. This is the cutting line $\mathbf{L}$ between the $\mathbf{xy}$ and **x'y'** planes. Here, α is the angle between $\mathbf{x}$ and $\mathbf{L}_\mathrm{a}$ and β is the angle between the two z–axes $\mathbf{z}$ and $\mathbf{z}'$. The angle γ is the angle between $\mathbf{L}_\mathrm{a}$ and $\mathbf{x}'$. The transformation of the CS requires three rotations in a specific order.

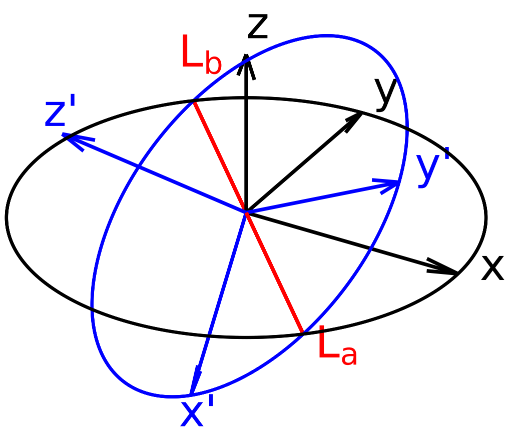

Figure 3.1: Depiction of two coordinate systems $\mathbf{x}, \mathbf{y}, \mathbf{z}$ (black) and $\mathbf{x}', \mathbf{y}', \mathbf{z}'$ (blue). The red cutting line of the $\mathbf{xy}$ and **x'y'** planes is between the vector $\mathbf{L}_\mathrm{a}$ and $\mathbf{L}_\mathrm{b}$.

For the rotation of the CS it is necessary to apply the rotation matrices in a specific order. The first rotation is around $\mathbf{z}$ by the angle γ, followed by the rotation $\mathbf{x}$ by β. Then a rotation around the $\mathbf{z}$ axis is done by the angle α. The rotation matrices $\mathbf{R}_z(\theta)$, and $\mathbf{R}_x(\theta)$ are mathematically described in Eq. (3.1.1) and Eq. (3.1.2), respectively. These are the rotation matrices for a classical rotation around the $\mathbf{z}$ and $\mathbf{c}$ axes.

$$\mathbf{R}_z(\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{3.1.1}$$

$$\mathbf{R}_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{pmatrix} \tag{3.1.2}$$

The full rotation of the coordinate system in dependency of the Euler angles is shown in Eq. (3.1.3).

$$\mathbf{R} = \mathbf{R}_z(\gamma) \times \mathbf{R}_x(\beta) \times \mathbf{R}_z(\alpha) \tag{3.1.3}$$

3.2 Electronic Structure Theory

In this report, properties derived from electronic structure theory are used for the training of the ML models. Although GFN2–xTB is used for this purpose, a short introduction to the Hartree–Fock model and Density Functional Theory is given. The introduction starts from the non–relativistic, stationary electronic Schrödinger–equation. From this the Hartree–Fock model will be derived. This is followed by Density Functional Theory, as this builds the theoretical basis of GFN2–xTB. After that the tight–binding method will be introduced from which the semiempirical method GFN2–xTB is derived. Atomic units are used for all equations in this section.

3.2.1 Hartree–Fock Model

Molecular systems in quantum mechanics can be described by a many–body wave function eigenvalue problem. The solution of this problem is given by solving the Schrödinger equation. The separation of its time–dependency, neglecting relativistic effects and the introduction of the Born–Oppenheimer (BO) approximation results in the electronic Schrödinger equation (Eq. (3.2.1)).

$$\hat{H}_\mathrm{e}\Psi_\mathrm{e} = E_\mathrm{e}\Psi_\mathrm{e} \tag{3.2.1}$$

The solution of this eigenvalue equation leads to the electronic energy E_e. The electronic many–body–wave function Ψ_e of the system can be described by the Slater determinant (Eq. (3.2.2)). The use of the Slater determinant fulfills the condition of the antisymmetry of the wave function. The condition is due to the Pauli exclusion principle. The Slater determinant consists of k spin–orbital wave functions ϕ_i, and every wave function is permuted with the coordinates $\mathbf{r}_i$ of the

N electrons. Furthermore, the wave function is normalized by the factor $(N!)^{-\frac{1}{2}}$.

$$\Psi_{\mathrm{e}} = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(\mathbf{r}_1) & \phi_2(\mathbf{r}_1) & \dots & \phi_k(\mathbf{r}_1) \\ \phi_1(\mathbf{r}_2) & \phi_2(\mathbf{r}_2) & & \vdots \\ \vdots & & \ddots & \vdots \\ \phi_1(\mathbf{r}_N) & \dots\dots & \dots & \phi_k(\mathbf{r}_N) \end{vmatrix} \tag{3.2.2}$$

The electronic Hamiltonian $\hat{H}_{\mathrm{e}}$ in the Hartree–Fock model is shown in Eq. (3.2.3) and consists of the kinetic energy of the electrons and the potential energy between electrons and nuclei as also between the electrons. Furthermore, the nuclear repulsion energy, which is approximated as a constant, is added. The kinetic energy of the nuclei can is separated, due to the BO–approximation.

$$\hat{H}_{\mathrm{e}} = \sum_{A=1}^{M} \sum_{i=1}^{N} \left(-\frac{1}{2}\nabla_i^2 - \frac{1}{|\mathbf{R}_A - \mathbf{r}_i|} \right) + \sum_{i=1}^{N} \sum_{\substack{j=1 \\ i>j}}^{N} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + \sum_{A=1}^{M} \sum_{\substack{B=1 \\ B>A}}^{M} \frac{Z_A Z_B}{|\mathbf{R}_A - \mathbf{R}_B|} \tag{3.2.3}$$

Here, $\mathbf{R}_A$ is the coordinates of atom A, $\mathbf{r}_i$ the coordinates of electron i and Z_A the nuclear charge. While, ∇_i^2 is the second derivative in respect to every coordinate of electron i.
The electronic energy can now be described with the Slater determinant and Hamiltonian in Eq. (3.2.4) This leads to the description of the energy by the one–electron operator h_i, the coulomb operator $\hat{J}_i$ and the exchange operator $\hat{K}$.

$$E_{\mathrm{e}} = \sum_{i=1}^{N} \langle \phi_i | h_i | \phi_i \rangle + \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \left(\langle \phi_j | \hat{J}_i | \phi_j \rangle - \langle \phi_j | \hat{K}_i | \phi_j \rangle \right) \tag{3.2.4}$$

The Rayleigh–Ritz variational principle (s. Eq. (3.2.5)) states, that the total energy E_0 of the system is always beneath or equal to the energy corresponding to the approximated wave function. This allows the search of the exact energy by optimizing the wave function (or its parameters).

$$E_0 \leq \langle \Psi | \hat{H} | \Psi \rangle \tag{3.2.5}$$

Hence, the determination of the total energy of the system is an eigenvalue problem and can be solved by minimization. To minimize the energy, Lagrangian optimization is employed, under the condition of orthonormal ϕ_i. The inner product of the wave function $\langle \phi_i | \phi_j \rangle = \delta_{ij}$ corresponds to the Kronecker delta. The Kronecker delta is $\delta_{ij} = 0$ if i and j are not equal, and $\delta_{ij} = 1$ if i and j are equal. The derivative of the Lagrangian functional leads to the canonical Fock equation (3.2.6), which is a solution for the minimal energy.

$$\hat{f}(\mathbf{r}_i)\phi_k(\mathbf{r}_i) = \epsilon(\mathbf{r}_i)\phi_k(\mathbf{r}_i) \tag{3.2.6}$$

$$\hat{f}(\mathbf{r}_i) = \hat{h} + \sum_{j=1}^{N}(2\hat{J}_j - \hat{K}_j) \tag{3.2.7}$$

The Fock–operator $\hat{f}(\mathbf{r}_i)$ is an effective one–electron operator. This enables the determination of the spin–orbital wave functions in an iterative manner, as the Fock–operator itself depends on the spin–orbital wave functions. The iterative search is possible due to the self–consistency of the Hartree–Fock equation. Furthermore, the general optimization can be achieved due to the Rayleigh–Ritz variational principle. Important to point out is that the Fock–operator consists of the one–electron operator, the exchange–operator and the coulomb operator, while the correlation is not included.

The spin–orbital wave functions are described via a basis set approximation. This LCAO–approach is shown in Eq. (3.2.8), where the spin–orbital wave–functions are the sum of K basis functions with an associated constant.

$$\phi_i(\mathbf{r}) = \sum_{\mu}^{K} c_\mu \chi_\mu(\mathbf{r}) \tag{3.2.8}$$

Merging the basis–set approach into Eq. (3.2.6) leads to Roothaan–Hall equation shown in Eq. (3.2.14). Here, $\mathbf{F}$ is the Fock matrix, $\mathbf{C}$ a matrix of coefficients c_μ, $\mathbf{S}$ the overlap matrix and $\boldsymbol{\epsilon}$ the diagonal eigenvalue matrix. This equation is used find a solution for the energy iteratively. Starting with an initial guess of the Fock matrix, the matrix $\mathbf{C}$ is computed, with this matrix, the density matrix $\mathbf{D}$ is calculated. From the matrix $\mathbf{D}$, is then used for the calculation of the new Fock matrix.

$$\mathbf{FC} = \mathbf{SC}\boldsymbol{\epsilon} \tag{3.2.9}$$

3.2.2 Kohn–Sham Density Functional Theory

As mentioned above, the correlation between electrons is not described in the Hartree–Fock model, due to the use of a single Slater determinant.

The correlation energy can be described by two different approaches. Instead of using one single Slater determinant, a linear combination of Slater determinants can be used. This leads to the post–Hartree–Fock methods like Full–CI or MP2. The second approach is the Density Functional Theory (DFT).

The Hohenberg–Kohn theorems build the basis for the DFT. These two theorems postulate, that the energy is a functional of the density, and that this functional follows the variational principle.

The description of the energy can therefore be done via the density of the electrons, which is defined in Eq. (3.2.10). Where, n_{occ} is the occupation number of the spin–orbital wave function.

$$\rho(\mathbf{r}) = n_{\text{occ}} \sum_{i=1}^{n} |\phi_i(\mathbf{r})|^2 \tag{3.2.10}$$

The general approach is to describe the energy in dependence of the electron density $E[\rho]$. There are different approaches for this connection. The most known approach, Kohn–Sham Density Functional Theory (KS–DFT), will be discussed. The energy in KS–DFT consists of the inexact kinetic energy, coulomb integral, the exchange correlation energy and the potential field. The kinetic energy is inexact, due to the assumption of non–interacting electrons. The energy E_{XC} consists mainly of the correlation energy and the exchange energy. In theory, the energy E_{XC} should also include the self interaction energy and a correction to the kinetic energy.

$$E[\{\phi_i\}] = T[\{\phi_i\}] + J[\rho] + E_{\text{XC}}[\rho] + \int v(\mathbf{r})\rho(\mathbf{r})\mathrm{d}\mathbf{r} \tag{3.2.11}$$

The minimization of the energy is done by a Lagrangian optimization with analogue conditions compared to Hartree–Fock. This leads to the Kohn–Sham operator and a new eigenequation, which is shown in Eq (3.2.12).

$$\hat{f}^{\text{KS}}(\mathbf{r}_i)\phi_k(\mathbf{r}_i) = \epsilon(\mathbf{r}_i)\phi_k(\mathbf{r}_i) \tag{3.2.12}$$

$$\hat{f}^{\text{KS}}(\mathbf{r}_i) = \hat{h}_i + 2\sum_{j=1}^{n} \hat{J}_j + \hat{v}_{\text{xc}} \tag{3.2.13}$$

In comparison to the Fock operator, $\hat{f}^{\text{KS}}(\mathbf{r}_i)$ includes an exchange–correlation term, instead of the exchange operator. With a basis set approach analogue to the HF model, an eigenvalue equation similar to the Roothaan–Hall equation is derived.

$$\mathbf{FC} = \mathbf{SC}\boldsymbol{\epsilon} \tag{3.2.14}$$

3.2.3 GFN2–xTB Electronic Structure Method

GFN2–xTB is a Density Functional Tight Binding (DFTB) method, that is optimized for **G**eomteries, **F**requencies and **N**oncovalent interaction energies. In these DFTB methods, the electron density ρ_0 is a linear combination of atomic electron densities $\rho_{0,A}$.

$$\rho_0 = \sum_{A} \rho_{0,A} \tag{3.2.15}$$

The energy is following expanded by a Taylor series around the electron density ρ_0. In GFN2–xTB the expansion is done up to the third order.

$$E[\rho] = E^{(0)}[\rho_0] + E^{(1)}[\rho_0, (\delta\rho)] + E^{(2)}[\rho_0, (\delta\rho)^2] + E^{(3)}[\rho_0, (\delta\rho)^3] + \dots \tag{3.2.16}$$

Furthermore, the energy is described with the kinetic energy $T[\rho(\mathbf{r})]$ and the external potential, due to the nuclei $V_n(\mathbf{r})$. A local density approximation (LDA) is used for the description of the exchange–correlation energy $\epsilon_{\mathrm{XC}}^{\mathrm{LDA}}$. As the LDA neglects the long range correlation effects, these are described via a non–local correlation contribution $\Phi_{\mathrm{C}}^{\mathrm{NL}}(\mathbf{r}, \mathbf{r}')$. This is in contrast to other DFTB methods like DTFB3.

$$\begin{aligned} E[\rho] = \int \rho(\mathbf{r}) \Big[& T[\rho(\mathbf{r})] + V_n(\mathbf{r}) + \epsilon_{\mathrm{XC}}^{\mathrm{LDA}}[\rho(\mathbf{r})] \\ & + \frac{1}{2} \int \left(\frac{1}{|\mathbf{r} - \mathbf{r}'|} + \Phi_{\mathrm{C}}^{\mathrm{NL}}(\mathbf{r}, \mathbf{r}') \right) \rho(\mathbf{r}') \, \mathrm{d}\mathbf{r}' \Big] \mathrm{d}\mathbf{r} + E_{nn} \end{aligned} \tag{3.2.17}$$

The terms of Eq. (3.2.16) can be divided into different energy contributions. These contributions will be briefly discussed. The zeroth order term shown in Eq. (3.2.18) consists of the repulsion energy $E_{\mathrm{rep}}^{(0)}$ and dispersion energy $E_{\mathrm{disp}}^{(0)}$. These are typically described by Buckingham or Lennard–Jones potentials. The dispersion term arises, due to the non–local correlation functional. Furthermore, the zeroth order term energy of the valence electrons $E_{\mathrm{A,valence}}^{(0)}$ and core electron $E_{\mathrm{A,core}}^{(0)}$ is summed, though the energy of the core electrons is defined as zero.

$$E[\rho_0] \approx E_{\mathrm{rep}}^{(0)} + E_{\mathrm{disp}}^{(0)} + \sum_A \left(E_{\mathrm{A,valence}}^{(0)} + E_{\mathrm{A,core}}^{(0)} \right) \tag{3.2.18}$$

The first order perturbation energy $E^{(1)}[\rho_0, (\delta\rho)]$ is approximately described by a first order dispersion energy $E_{\mathrm{disp}}^{(1)}$, and first order valence electron energy $E_{\mathrm{A,valence}}^{(1)}$. This perturbation allows the description of covalent bonding, as the electron density so far is described as spherical densities around the atoms, with no interaction.

$$E^{(1)}[\rho_0, (\delta\rho)] \approx E_{\mathrm{disp}}^{(1)} + \sum_A E_{\mathrm{A,valence}}^{(1)} \tag{3.2.19}$$

The second order energy perturbation $E^{(2)}[\rho_0, (\delta\rho)^2]$ is described by the contribution of dispersion, electrostatics and exchange correlation energies. The exchange and correlation energy $E_{\mathrm{XC}}^{(2)}$ and electrostatic energy $E_{\mathrm{ES}}^{(2)}$ are described in GFN2–xTB with an isotropic as also an anisotropic contribution. This is in contrast to other DFTB methods, where only a monopole approximation is done. Hence, there is only an isotropic description in other methods.

$$E^{(2)}[\rho_0,(\delta\rho)^2] \approx E^{(2)}_{\mathrm{ES}} + E^{(2)}_{\mathrm{XC}} + E^{(2)}_{\mathrm{disp}} \tag{3.2.20}$$

For the third order perturbation, only the electrostatic and exchange correlation energies are used and computed with the Mulliken approximation. All other contributions to the third order perturbation are neglected.

$$E^{(3)}[\rho_0,(\delta\rho)^3] \approx E^{(3)}_{\mathrm{ES}} + E^{(3)}_{\mathrm{XC}} \tag{3.2.21}$$

Merging all energy contributions shown in Eq. (3.2.18) to Eq. (3.2.21) into Eq. (3.2.16) leads to Eq. (3.2.22). Though, some further adjustments are done. The isotropic contribution of the electrostatic and exchange correlation energies are described by a single term $E_{\mathrm{IES+IEC}}$. While the anisotropic electrostatic energy E_{AES} and anisotropic exchange correlation energy E_{AXC} are described separately. The valence electron energy contributions are described via the extended Hückel type energy E_{EHT}. Independent of the above discussed perturbation, an energy term G_{Fermi} describing the Fermi–smearing is included, which leads to the complete energy equation shown in Eq. (3.2.22).

$$E_{\mathrm{GFN2-xTB}} = E_{\mathrm{rep}} + E_{\mathrm{EHT}} + E_{\mathrm{disp}} + E_{\mathrm{IES+IEC}} + E_{\mathrm{AES}} + E_{\mathrm{AXC}} + G_{\mathrm{Fermi}} \tag{3.2.22}$$

The wave function is approximated by a minimal valence basis set STO–mG. The energy is computed similar to KS–DFT and Hartree Fock. The Roothaan–Hall type Eq. (3.2.14) is solved in a selfconsistent manner.

3.3 Hessian Matrices

Hessian matrices are required for several uses like the computation of wave numbers or thermodynamic quantities. Furthermore, they are necessary for the transition state search. The hessian matrix is a $3M \times 3M$ matrix of second derivatives of the energy in respect to every permuted atom coordinate pair in the system.

$$\mathbf{H} = \begin{pmatrix} \frac{\partial^2 E}{\partial x_A \partial x_A} & \cdots & \frac{\partial^2 E}{\partial x_A \partial y_M} & \frac{\partial^2 E}{\partial x_A \partial z_M} \\ \frac{\partial^2 E}{\partial x_A \partial y_A} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\partial^2 E}{\partial z_N \partial x_A} & \cdots & & \frac{\partial^2 E}{\partial z_M \partial z_M} \end{pmatrix} \tag{3.3.1}$$

The hessian matrix can be divided into M^2 3×3 block matrices, where each depends on the coordinates of an atom pair. Theses 3×3 block matrices are beneficial for the prediction, as they only depend on two atoms. Hence, only information about the two atoms need to be given to the machine learning model to predict the block matrix.

$$\mathbf{H} = \begin{pmatrix} \mathbf{H}_{AA} & \mathbf{H}_{AB} & \dots & \mathbf{H}_{AM} \\ \mathbf{H}_{BA} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \mathbf{H}_{MA} & \dots\dots & \dots & \mathbf{H}_{MM} \end{pmatrix} \tag{3.3.2}$$

Due to the Schwartz Theorem, the $\mathbf{H}_{BA}$ and $\mathbf{H}_{AB}$ are related to another by Eq. (3.3.3)

$$\mathbf{H}_{BA} = (\mathbf{H}_{AB})^{\intercal} \tag{3.3.3}$$

Therefore, the only information needed for the hessian is a triangular matrix (s. Eq. (3.3.4)).

$$\mathbf{H} = \begin{pmatrix} \mathbf{H}_{AA} & \mathbf{H}_{AB} & \dots & \mathbf{H}_{AM} \\ & \ddots & & \vdots \\ & \text{\huge 0} & \ddots & \vdots \\ & & & \mathbf{H}_{MM} \end{pmatrix} \tag{3.3.4}$$

The hessian matrix needs to be projected for the computation of physical quantities like the harmonic frequencies, zero point vibrational energy. For the projection the rotational and translational parts of the hessian need to be detected. A $6 \times 3N$ matrix is constructed, that describes the rotation and translation of each atom. For one atom, the overlap is depicted in Eq. (3.3.5). The translation correspond to the first 3 rows of the matrix $\mathbf{O}_A$. While the contribution of the rotation is given in the lower 3 rows. For example, the rotation around the x–axis leads to the translation of the y–coordinate $\mathbf{r}_A^y$ by $\mathbf{r}_A^z$. While the z–coordinate is moved by $-\mathbf{r}_A^y$.

$$\mathbf{O}_A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & \mathbf{r}_A^z & -\mathbf{r}_A^y \\ -\mathbf{r}_A^z & 0 & \mathbf{r}_A^x \\ \mathbf{r}_A^y & -\mathbf{r}_A^x & 0 \end{pmatrix} \tag{3.3.5}$$

For a system of N atoms, the full overlap matrix is shown in Eq. (3.3.6)

$$\mathbf{O} = \begin{pmatrix} \mathbf{O}_A & \mathbf{O}_B & \dots & \mathbf{O}_N \end{pmatrix} \tag{3.3.6}$$

After that, the rotation and translation matrix is normalized to $\mathbf{O}_n$. For the detection, the overlap matrix $\mathbf{M}$ between the normalized rotation and translation matrix and the hessian matrix is computed (s. Eq. (3.3.7)). The multiplication of the $6 \times 3N$ and $3N \times 3N$ matrix leads to $6 \times 3N$ matrix.

$$\mathbf{M} = \mathbf{O} \times \mathbf{H} \tag{3.3.7}$$

The columns of $\mathbf{M}$ are summed up, which leads to a $3N$ vector. The maximum values of this vector corresponds to the highest overlap between a row in the Hessian matrix and the translations and rotation matrix. The six highest values are searched, and the numbers of the corresponding rows are stored in a vector $\mathbf{v}$.
For a linear molecule only the five highest values are searched. The differentiation between linear and non–linear molecules is done by the norm of the coordinates $\|\mathbf{r}^x\|$,$\|\mathbf{r}^y\|$ and $\|\mathbf{r}^z\|$. If the sum of the norm of two coordinates is below a threshold of $1 \cdot 10^{-6}$ a linear molecule is assumed.
For the projection itself, the eigenvectors $\mathbf{Q}$ of the Hessian matrix and its eigenvalues $\boldsymbol{\lambda}$ are required.

$$\mathbf{H}^{\text{proj}} = \mathbf{H} - \sum_{i \in \mathbf{v}} \boldsymbol{\lambda}_i \cdot \mathbf{Q}^{\intercal}{}_i \otimes (\mathbf{Q}^{\intercal}{}_i)^{\intercal} \tag{3.3.8}$$

At last, the hessian needs to be mass weighted. Every element is multiplied by a specifc factor, which depends on the mass m_A of the atoms A of the coordinates.

$$\mathbf{H}^{\text{proj,w}}_{AB} = \frac{1}{\sqrt{m_A m_B}} \mathbf{H}^{\text{proj}}_{AB} \tag{3.3.9}$$

Eq. (3.3.9) is applied to every hessian matrix block $\mathbf{H}_{AB}$ in the full Hessian matrix $\mathbf{H}$, which leads to the projected mass weighted Hessian matrix $\mathbf{H}^{\text{proj,w}}$.

3.4 Machine Learning

Extra Tree Regression (ETR) and Random Forest Regression (RFR) are used as ML algorithms in this research. Both algorithms are ensemble methods based on decision trees as individual estimators. In this section, decision tree based algorithms will be discussed, and the differences between the ETR and RFR will be emphasized.
Often these decision tree based algorithms are based on the Classification and Regression Tree

(CART) algorithm. For an example, a response Y that depends on two variables $X1$ and $X2$ is assumed. The variables $X1$ and $X2$ are called features. A tree based model splits the space of this function into several recursive binary partitions. Recursive binary means in this case, that the splits are orthogonal to the axis of the features. This split is shown on the left in Fig. 3.2.

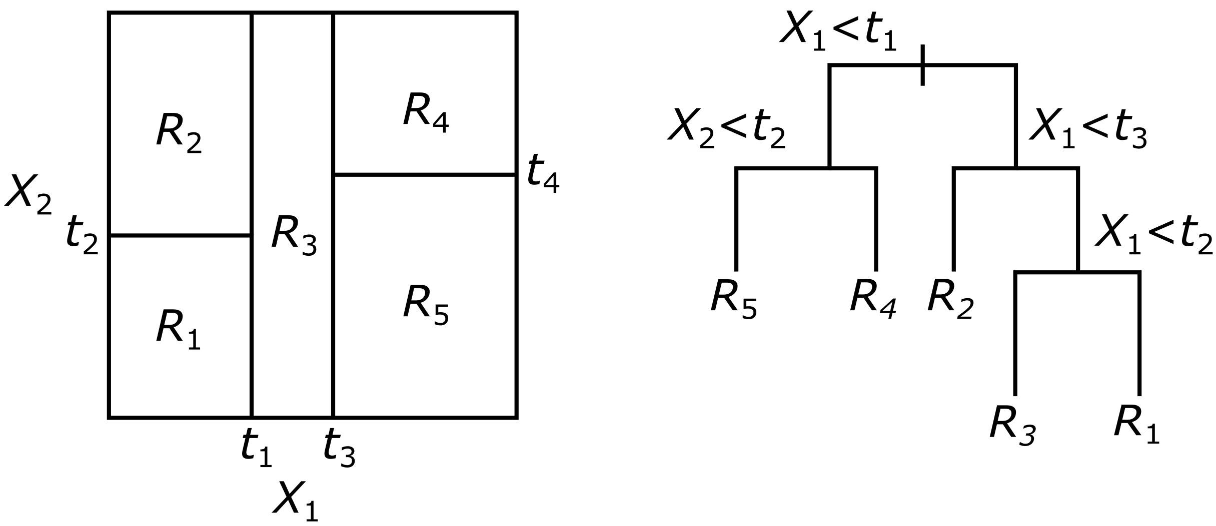

Figure 3.2: The recursively binary grown split is shown on the left. On the right, the resulting Decision Tree is depicted.

In each partition Y is modeled with a function or constant R_i. The splitting points t_1 to t_4 are chosen differently for the RFR and ETR. In RFR, the splitting point is determined by choosing the best feature and of this feature the best split value. Hence, it is a two dimensional search. In the case of the ETR, the feature is chosen randomly. After that, the best split value for this feature is chosen. The relation between the features and the partitioned response Y can be depicted in a decision tree, which is shown in Fig. 3.2 on the right. This can also be described mathematically via Eq. (3.4.1)

$$\hat{f}(X) = \sum_{m=1}^{5} c_m I\{(X_1, X_2) \in R_m\} \tag{3.4.1}$$

Here, $\hat{f}(X)$ is the function that describes the response Y. Each region m has a corresponding constant c_m. The function I can either be $I = 1$, if the features X_1 and X_2 are in the rectangle R_m, or it is $I = 0$, if it is not the case. Originally, the ETR and RFR have another difference, which is that the ETR did not use bootstrapping. Bootstrapping uses the training set and makes several new training sets from this. The new training sets are randomly filled with the datasets of the original training set.

4 Methods

5 Results and Discussion

6 Conclusion and Outlook

7 Bibliography